Jack Ao

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EDUCATION

University of Dayton Expected May 2030
Bachelor of Science in Mechanical Engineering Dayton, OH

- *Relevant Coursework*: Calculus I, Physics I (Mechanics), Introduction to Engineering Design, Engineering Graphics/CAD, Introduction to Python

Benet Academy Aug. 2022 – June 2026
High School Diploma Lisle, IL

- *Relevant Coursework*: Calculus I, Physics (Mechanics, Electricity & Magnetism), Principles of Engineering, Introduction to Aerospace Engineering, Basic Python

EXTRACURRICULAR ACTIVITIES

Aero Design Team (AIAA) — University of Dayton Sept. 2026 – Present
Member, Aerodynamics Team Dayton, OH

- Collected and analyzed data from past competition designs in spreadsheets to compare wing loading, weight, and mission scores, in order to identify the optimal aircraft configuration
- Modeled airfoils and wing configurations in XFLR5, comparing lift, drag, and stability results to select the design for a UAV that carries, deploys, tows, and recovers a sensor across three scored missions
- Collaborated with the structural design team, sharing aerodynamic results and iterating wing geometry against weight and strength constraints so the design stayed both efficient and buildable
- Support the surveillance missions by deploying the towed sensor in flight, holding a stable tow at altitude, and recovering it cleanly before landing
- Carry assigned components from CAD through fabrication, ground testing, and flight testing

PROJECTS

Wood Plane Jan. 2026 – March 2026

- Compared airfoils in Aery32 and selected the section with the best lift-to-drag behavior for a hand-launched glider
- Created a model of each part of the aircraft — fuselage, wing, ailerons, and stabilizers — before moving to fabrication
- Hand-measured, cut, and assembled every part from stock, sanding and aligning surfaces to keep the airframe true
- Achieved a flight distance of 71 feet with a 30-gram mass at the nose, tuning nose weight to trim the aircraft for a stable glide

SKILLS

Design & Simulation : SolidWorks, AutoCAD, XFLR5, Aery32, OpenRocket
Programming : Python
Fabrication : 3D printing, hand fabrication and assembly, basic shop tools
Technical : Engineering graphics, technical drawing, CAD modeling, airfoil selection